

Name: Brittany 

Date: June 18, 2025

Instructor: Samuel Chukwuemeka

Project: Inferential Statistics

First Dataset

Inferential Statistics: Population Proportion: One Sample

Dataset: Nutrition, Physical Activity, and Obesity - American Community Survey

Source: Data.gov: The Home of the U.S. Government's Open Data

General Description: (Taken from: <https://catalog.data.gov/dataset/nutrition-physical-activity-and-obesity-american-community-survey>)

This dataset includes select data from the U.S. Census Bureau's American Community Survey (ACS) on the percent of adults who bike or walk to work. This data is used for DNPAO's Data, Trends, and Maps database, which provides national and state specific data on obesity, nutrition, physical activity, and breastfeeding.

Specific Description:

The data set Nutrition, Physical Activity, and Obesity - American Community Survey.

Specifically, we are interested in determining the proportion of estimating the proportion of **U.S. adults who usually bike or walk to work in the last week.**

So, in the file, we look at the **ACS_physical_activity.csv** page and column **H. and K.**

Save As

Recent

OneDrive - Personal

Desktop

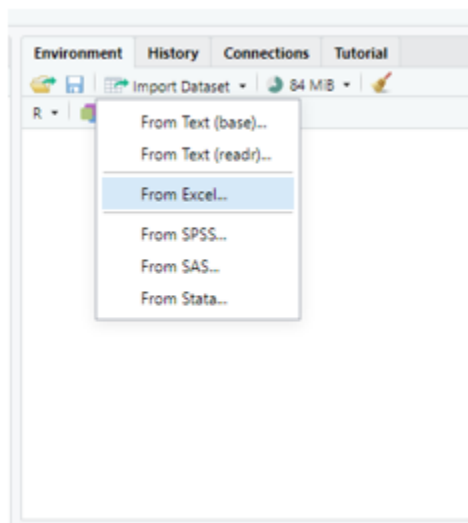
ACS_physical_activity.csv

CSV (Comma delimited) (*.csv)

Save

1st

2nd



Import Excel Data

File/URL:

Browse...

Data Preview:

Import Options:

Name: dataset Max Rows: First Row as Names

Sheet: Default Skip: 0 Open Data Viewer

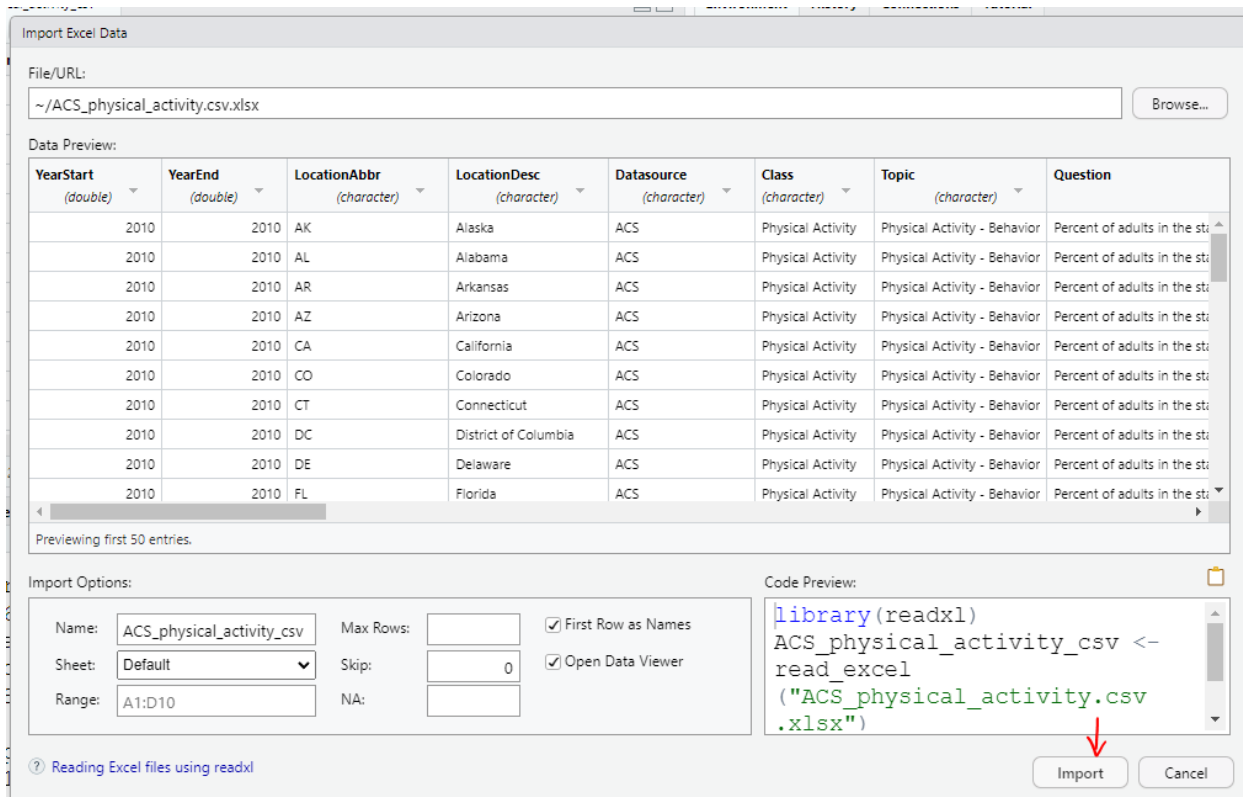
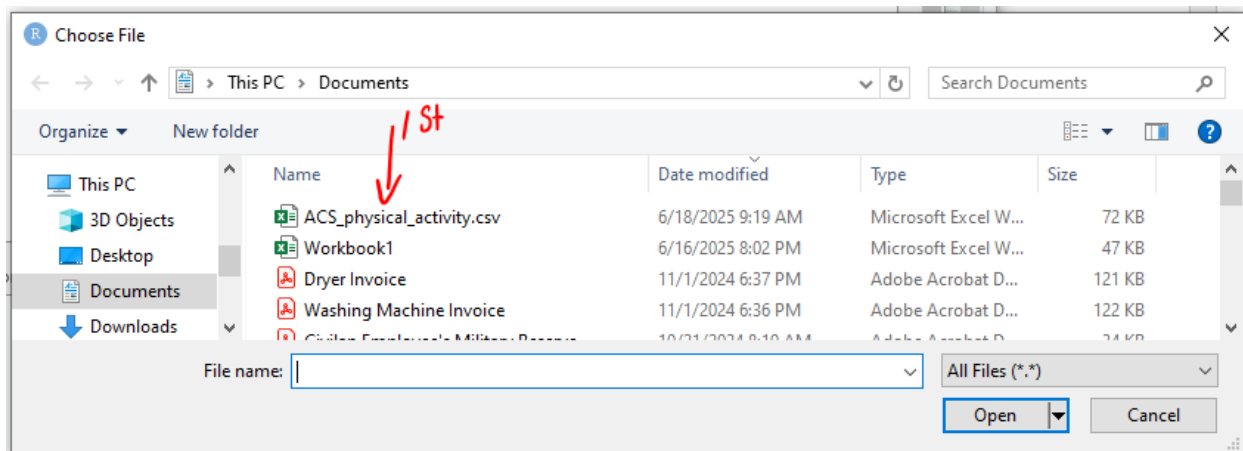
Range: A1:D10 NA:

Code Preview:

```
library(readxl)
dataset <- read_excel(NULL)
View(dataset)
```

Reading Excel files using readxl

Import Cancel



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Project: (None)

Environment History Connections Tutorial

R Global Environment 131 MiB

Import Dataset

Data

ACS_physic... 636 obs. of 27 variables

Files Plots Packages Help Viewer Presentation

Zoom Export

Console Terminal Background Jobs

```
R - R 4.5.1 - ~/
> library(readxl)
> ACS_physical_activity_csv <- read_excel("ACS_physical_activity.csv.xlsx")
> View(ACS_physical_activity_csv)
> |
```

	YearStart	YearEnd	LocationAbbr	LocationDesc	Datasource	Class	Topic
1	2010	2010	AK	Alaska	ACS	Physical Activity	Physical
2	2010	2010	AL	Alabama	ACS	Physical Activity	Physical
3	2010	2010	AR	Arkansas	ACS	Physical Activity	Physical
4	2010	2010	AZ	Arizona	ACS	Physical Activity	Physical
5	2010	2010	CA	California	ACS	Physical Activity	Physical
6	2010	2010	CO	Colorado	ACS	Physical Activity	Physical
7	2010	2010	CT	Connecticut	ACS	Physical Activity	Physical
8	2010	2010	DC	District of Columbia	ACS	Physical Activity	Physical
9	2010	2010	DE	Delaware	ACS	Physical Activity	Physical
10	2010	2010	FL	Florida	ACS	Physical Activity	Physical
11	2010	2010	GA	Georgia	ACS	Physical Activity	Physical
12	2010	2010	HI	Hawaii	ACS	Physical Activity	Physical

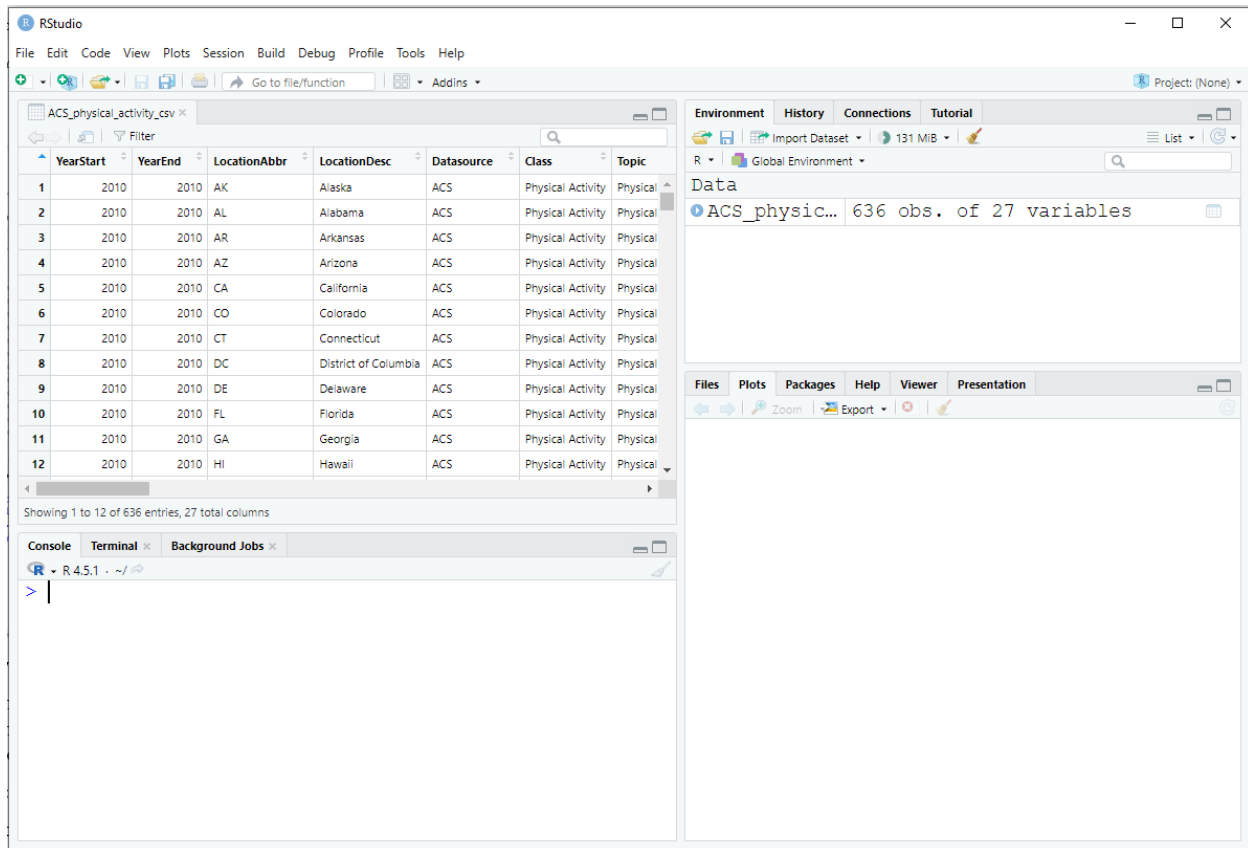
Showing 1 to 12 of 636 entries, 27 total columns

Let us clear the console window so we can have more space to write our code

The screenshot shows the RStudio interface. The top-left pane displays a data table with columns: YearStart, YearEnd, LocationAbbr, LocationDesc, Datasource, Class, and Topic. The table contains 12 rows of data for the year 2010, covering various US states and the District of Columbia. The top-right pane shows the Environment tab with a data object named 'ACS_physic...' containing 636 observations of 27 variables. The bottom-left pane shows the Console window with the following R code:

```
> library(readxl)
> ACS_physical_activity_csv <- read_excel("ACS_physical_activity.csv.xlsx")
> View(ACS_physical_activity_csv)
> |
```

A red arrow points to the 'View' function in the code, indicating the next step in the process.



We are only interested in:

- (a.) The percentage of adults who usually biked or walked to work (Column: Data_Value)

These are numeric percentages values, and we convert them to proportions by dividing by 100.

- (b.) The overall adult population, not segmented by gender, age, or other subgroups. (Column: StratificationCategory1)

The rows where StratificationCategory1 is “Total.”

Objectives:

- (1.) Construct a 95% confidence interval for the population proportion of U.S. adults who usually biked or walked to work in the last week.
- (2.) Interpret the confidence interval.
- (3.) Estimate the population proportion of U.S. adults who usually biked or walked to work in the last week.
- (4.) Conduct a hypothesis test to determine whether the population proportion of U.S. adults who usually biked or walked to work differs significantly from 5%.

Parameter to estimate: Population Proportion

Test: *prop-test*

Reason for Test: The population standard deviation was not given, and we are working with proportions.

Verify Requirements for Test

- (1.) The sample is a simple random sample (assumed, as this is national/state-level CDC survey data).
- (2.) The sampling distribution of the sample proportion is approximately normal, because

$$\hat{p} = \frac{x}{n}$$

$$n\hat{p} \geq 5$$

$$n\hat{q} \geq 5$$

Where:

$\hat{p}$ = sample proportion or estimated proportion of successes

$\hat{q}$ = sample proportion or estimated proportion of failures

x = number of individuals in the sample with the specified characteristics

n = sample size

Given a large, assumed sample size (e.g., 100,000), these conditions are easily satisfied.

Write the codes to complete the following tasks:

- (a.) Determine the number of adults who usually walk or bike to work in the last week.

This is the numerator: x (Data_Value)

It is the number of individuals in the sample with the specified characteristic.

- (b.) Determine the total number of individuals in the assumed sample.

This is the denominator: n

This is the assumed sample size ($n = 100,000$).

Let us begin by writing the codes for RStudio:

- (1.) Load the `dplyr` package for filtering.

```
Console Terminal x Background Jobs x
R 4.5.1 ~/
> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
  filter, lag

The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union

> |
```

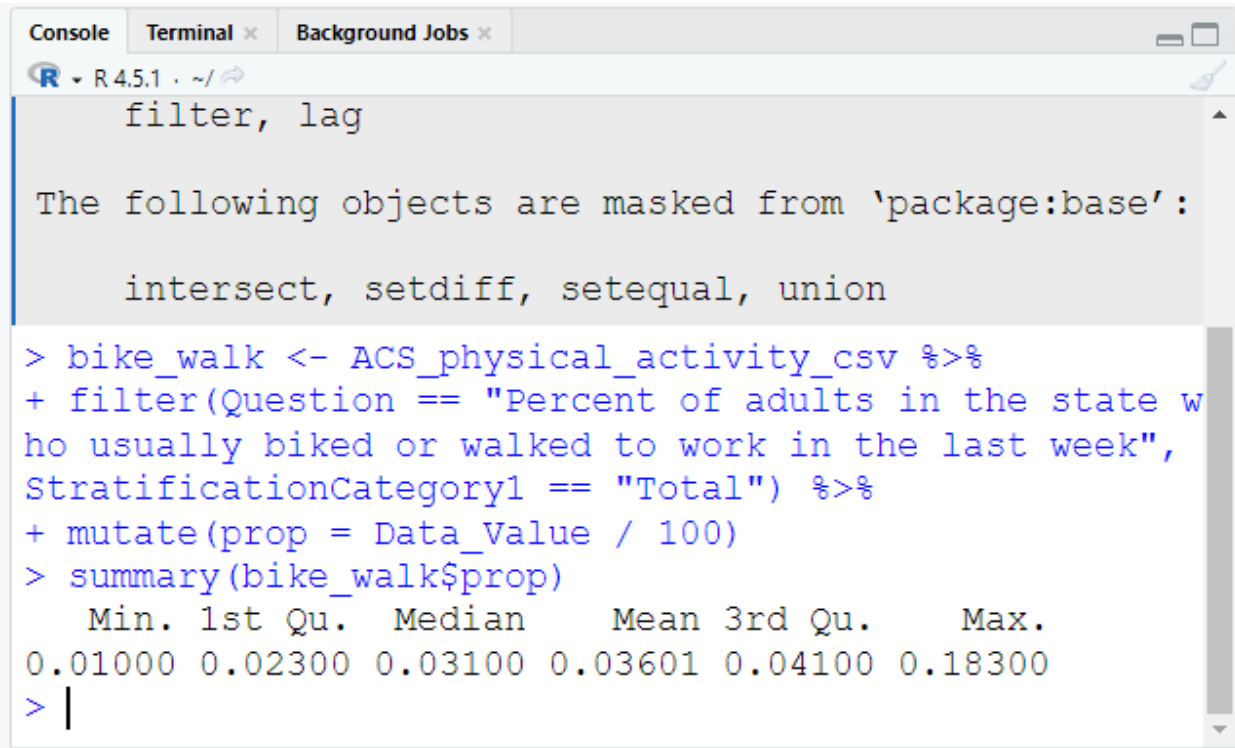
(2.) Filter and create proportion variable.

```
Console Terminal x Background Jobs x
R 4.5.1 ~/
The following objects are masked from 'package:stats':
  filter, lag

The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union

> bike_walk <- ACS_physical_activity_csv %>%
+ filter(Question == "Percent of adults in the state w
ho usually biked or walked to work in the last week",
StratificationCategory1 == "Total") %>%
+ mutate(prop = Data_Value / 100)
> |
```

(3.) Preview the cleaned data.



```
Console Terminal x Background Jobs x
R 4.5.1 ~ /
filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

> bike_walk <- ACS_physical_activity_csv %>%
+ filter(Question == "Percent of adults in the state w
ho usually biked or walked to work in the last week",
StratificationCategory1 == "Total") %>%
+ mutate(prop = Data_Value / 100)
> summary(bike_walk$prop)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.01000 0.02300 0.03100 0.03601 0.04100 0.18300
> |
```

NOTE: The next steps will estimate and test whether the proportion of U.S. adults who walk or bike to work is significantly different from a national benchmark of 5% (0.05).

(4.) Compute the sample proportion.

```

Console Terminal Background Jobs
R 4.5.1 ~ /

intersect, setdiff, setequal, union

> bike_walk <- ACS_physical_activity_csv %>%
+ filter(Question == "Percent of adults in the state w
ho usually biked or walked to work in the last week",
StratificationCategory1 == "Total") %>%
+ mutate(prop = Data_Value / 100)
> summary(bike_walk$prop)
  Min. 1st Qu.  Median    Mean 3rd Qu.   Max.
0.01000 0.02300 0.03100 0.03601 0.04100 0.18300
> phat <- mean(bike_walk$prop, na.rm = TRUE)
> phat
[1] 0.03600943
> |

```

(5.) Assume a sample size (e.g., 100,000)

The screenshot shows the RStudio interface. The top-left pane displays a data table with columns: YearStart, YearEnd, LocationAbbr, LocationDesc, Datasource, Class, and Topic. The table lists data for various US states from 2010 to 2011, sourced from the ACS, categorized under Physical Activity. The bottom-left pane shows the R console with the following code and output:

```

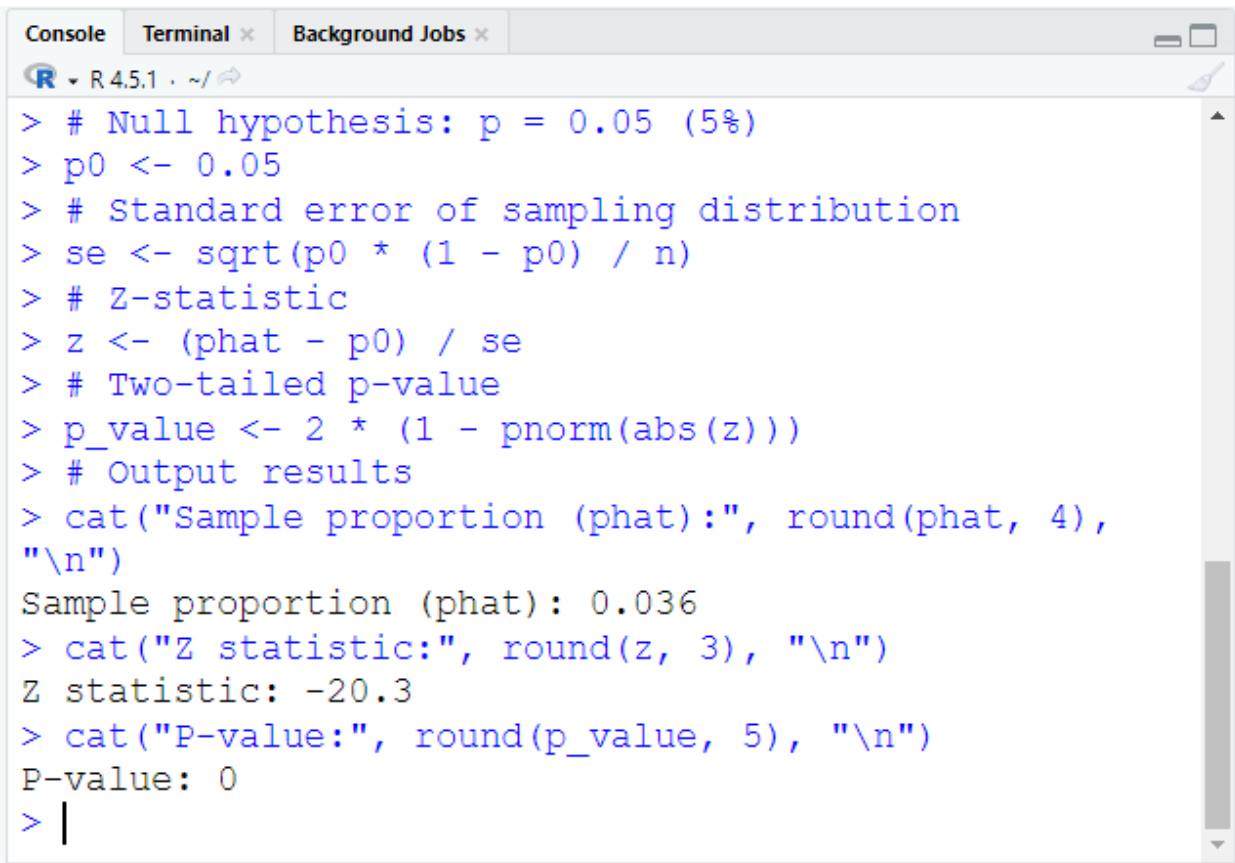
R 4.5.1 ~ /
filter(Question == "Percent of adults in the state w
ho usually biked or walked to work in the last week",
StratificationCategory1 == "Total") %>%
+ mutate(prop = Data_Value / 100)
> summary(bike_walk$prop)
  Min. 1st Qu.  Median    Mean 3rd Qu.   Max.
0.01000 0.02300 0.03100 0.03601 0.04100 0.18300
> phat <- mean(bike_walk$prop, na.rm = TRUE)
> phat
[1] 0.03600943
> n <- 100000 # Assumed total sample size (large survey
assumption)
> x <- round(phat * n) # Estimated number of people wh
o walk or bike
> |

```

The top-right pane shows the Environment window with two objects: 'ACS_physic...' (636 obs. of 27 variables) and 'bike_walk' (636 obs. of 28 variables). The bottom-right pane shows the Values window with the following data:

Values	
n	1e+05
phat	0.0360094339622642
x	3601

(6.) Run Z-test for proportions.

A screenshot of an R console window. The window has three tabs: 'Console', 'Terminal', and 'Background Jobs'. The 'Console' tab is active. The R version is 4.5.1. The code entered in the console is as follows:

```
> # Null hypothesis: p = 0.05 (5%)
> p0 <- 0.05
> # Standard error of sampling distribution
> se <- sqrt(p0 * (1 - p0) / n)
> # Z-statistic
> z <- (phat - p0) / se
> # Two-tailed p-value
> p_value <- 2 * (1 - pnorm(abs(z)))
> # Output results
> cat("Sample proportion (phat):", round(phat, 4),
"\n")
Sample proportion (phat): 0.036
> cat("Z statistic:", round(z, 3), "\n")
Z statistic: -20.3
> cat("P-value:", round(p_value, 5), "\n")
P-value: 0
> |
```

(7.) Confirm with `prop.test` (alternative method)

```
Console Terminal x Background Jobs x
R 4.5.1 ~ /
> prop.test(x = x, n = n, p = 0.05, alternative = "two.sided", correct = TRUE)

      1-sample proportions test with continuity
      correction

data:  x out of n, null probability 0.05
X-squared = 411.75, df = 1, p-value <
2.2e-16
alternative hypothesis: true p is not equal to 0.05
95 percent confidence interval:
 0.03486801 0.03718779
sample estimates:
           p
0.03601

> |
```

Based on the results:

The sample proportion of U.S. adults who usually biked or walked to work in the last week is approximately **3.6%**. A 95% confidence interval for the true population proportion is:

[0.03486801, 0.03718779]

This means we are 95% confident that the true proportion of all U.S. adults who usually biked or walked to work falls between **3.49% and 3.72%**.

We also conducted a hypothesis test to determine whether the true population proportion differs from **5%**. The null hypothesis was:

$$H_0: p = 0.05$$

The alternative hypothesis was:

$$H_0: p \neq 0.05$$

The test yielded a chi-squared value of **411.75** with a **p-value** $< 2.2\text{e-}16$, which is extremely small. Since the p-value is much less than 0.05, we **reject the null hypothesis**.

Second Dataset

Inferential Statistics: Population Mean: One Sample

Dataset: Sub-hourly stream temperature measurements throughout Delaware Water Gap National Recreation Area and New River Gorge National Park and Preserve (2022-2023) Data Package

Source: Data.gov: The Home of the U.S. Government's Open Data

General Description: (Taken from: <https://catalog.data.gov/dataset/sub-hourly-stream-temperature-measurements-throughout-delaware-water-gap-national-recreati>)

Continuous water temperature measurements were recorded at 59 locations in wadeable streams (watershed size = 2 - 100 km²) throughout Delaware Water Gap National Recreation Area and New River Gorge National Park and Preserve from April 2022 to November 2023. Data loggers were deployed at probabilistically selected sites where benthic macroinvertebrate, fish, and streamside bird communities have been monitored by the Eastern Rivers and Mountains Network (National Park Service) since 2014.

Variable: Water Temperature

Unit of the variable: Degrees of Celsius (°C)

Sample size: 94,827

Sample: All readings during July–August 2023

Population: Entire set of possible temperature readings during July-August 2023.

Year: 2023

Objectives:

- (1.) Construct a 95% confidence interval for the population mean of stream water temperatures at the Delaware Water Gap National Recreation Area during July and August 2023.
- (2.) Interpret the confidence interval.
- (3.) Estimate the population mean of stream water temperatures at the Delaware Water Gap National Recreation Area during July and August 2023.
- (4.) Test the claim that the population mean stream temperature during this period is equal to 18 °C using a one-sample t-test.
- (5.) Interpret the results of the hypothesis test in the context of environmental thresholds for cold-water aquatic species.

Parameter to estimate: Population Mean

Test: *t-test*

Reason for Test: The population standard deviation was not given.

Verify Requirements for Test

(1.) The sample is a simple random sample.

The dataset includes time-based sensor readings from stream monitoring stations. While not randomly selected in a strict statistical sense, the data represent a systematic and unbiased sampling process over time. This is an acceptable assumption for inference.

(2.) The population is normally distributed.

With a large sample size ($n > 30$), the Central Limit Theorem ensures that the sampling distribution of the sample mean is approximately normal, even if the population is not perfectly normal.

(3.) The sample size is less than 5% of the population size.

The sample (sub-hourly readings over two months) represents a small fraction of the theoretically infinite possible stream temperature readings, satisfying this condition.

Specifically, we are interested in testing whether the average stream water temperature during summer 2023 exceeded 18 °C.

So, in the file, we look at the **ERMN_DataLogger.csv** page and column **D.** and **F.**

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat

Paste Cut Copy Format Painter Clipboard Font Alignment Number

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in

A1 SiteID

	A	B	C	D	E	F	G	H	I	J
1	SiteID		SiteName	DateTime	Variable	Value	Units	Grade	Latitude	Longitude
2	ERMN_DEI DEWA	AdamsCre	2022-04-12T19:00:00Z	Water Temp	11.516	degC	Excellent	41.24803	-74.8769	
3	ERMN_DEI DEWA	AdamsCre	2022-04-12T20:00:00Z	Water Temp	11.88	degC	Excellent	41.24803	-74.8769	
4	ERMN_DEI DEWA	AdamsCre	2022-04-12T21:00:00Z	Water Temp	12.05	degC	Excellent	41.24803	-74.8769	
5	ERMN_DEI DEWA	AdamsCre	2022-04-12T22:00:00Z	Water Temp	12.195	degC	Excellent	41.24803	-74.8769	
6	ERMN_DEI DEWA	AdamsCre	2022-04-12T23:00:00Z	Water Temp	12.219	degC	Excellent	41.24803	-74.8769	
7	ERMN_DEI DEWA	AdamsCre	2022-04-13T00:00:00Z	Water Temp	12.098	degC	Excellent	41.24803	-74.8769	
8	ERMN_DEI DEWA	AdamsCre	2022-04-13T01:00:00Z	Water Temp	11.832	degC	Excellent	41.24803	-74.8769	
9	ERMN_DEI DEWA	AdamsCre	2022-04-13T02:00:00Z	Water Temp	11.467	degC	Excellent	41.24803	-74.8769	
10	ERMN_DEI DEWA	AdamsCre	2022-04-13T03:00:00Z	Water Temp	11.102	degC	Excellent	41.24803	-74.8769	
11	ERMN_DEI DEWA	AdamsCre	2022-04-13T04:00:00Z	Water Temp	10.761	degC	Excellent	41.24803	-74.8769	
12	ERMN_DEI DEWA	AdamsCre	2022-04-13T05:00:00Z	Water Temp	10.418	degC	Excellent	41.24803	-74.8769	
13	ERMN_DEI DEWA	AdamsCre	2022-04-13T06:00:00Z	Water Temp	10.1	degC	Excellent	41.24803	-74.8769	
14	ERMN_DEI DEWA	AdamsCre	2022-04-13T07:00:00Z	Water Temp	9.829	degC	Excellent	41.24803	-74.8769	
15	ERMN_DEI DEWA	AdamsCre	2022-04-13T08:00:00Z	Water Temp	9.583	degC	Excellent	41.24803	-74.8769	
16	ERMN_DEI DEWA	AdamsCre	2022-04-13T09:00:00Z	Water Temp	9.435	degC	Excellent	41.24803	-74.8769	
17	ERMN_DEI DEWA	AdamsCre	2022-04-13T10:00:00Z	Water Temp	9.336	degC	Excellent	41.24803	-74.8769	
18	ERMN_DEI DEWA	AdamsCre	2022-04-13T11:00:00Z	Water Temp	9.262	degC	Excellent	41.24803	-74.8769	
19	ERMN_DEI DEWA	AdamsCre	2022-04-13T12:00:00Z	Water Temp	9.312	degC	Excellent	41.24803	-74.8769	
20	ERMN_DEI DEWA	AdamsCre	2022-04-13T13:00:00Z	Water Temp	9.386	degC	Excellent	41.24803	-74.8769	
21	ERMN_DEI DEWA	AdamsCre	2022-04-13T14:00:00Z	Water Temp	9.682	degC	Excellent	41.24803	-74.8769	
22	ERMN_DEI DEWA	AdamsCre	2022-04-13T15:00:00Z	Water Temp	10.124	degC	Excellent	41.24803	-74.8769	
23	ERMN_DEI DEWA	AdamsCre	2022-04-13T16:00:00Z	Water Temp	10.98	degC	Excellent	41.24803	-74.8769	
24	ERMN_DEI DEWA	AdamsCre	2022-04-13T17:00:00Z	Water Temp	12.001	degC	Excellent	41.24803	-74.8769	
25	ERMN_DEI DEWA	AdamsCre	2022-04-13T18:00:00Z	Water Temp	12.727	degC	Excellent	41.24803	-74.8769	
26	ERMN_DEI DEWA	AdamsCre	2022-04-13T19:00:00Z	Water Temp	13.161	degC	Excellent	41.24803	-74.8769	
27	ERMN_DEI DEWA	AdamsCre	2022-04-13T20:00:00Z	Water Temp	13.618	degC	Excellent	41.24803	-74.8769	
28	ERMN_DEI DEWA	AdamsCre	2022-04-13T21:00:00Z	Water Temp	13.93	degC	Excellent	41.24803	-74.8769	
29	ERMN_DEI DEWA	AdamsCre	2022-04-13T22:00:00Z	Water Temp	14.122	degC	Excellent	41.24803	-74.8769	
30	ERMN_DEI DEWA	AdamsCre	2022-04-13T23:00:00Z	Water Temp	14.146	degC	Excellent	41.24803	-74.8769	
31	ERMN_DEI DEWA	AdamsCre	2022-04-14T00:00:00Z	Water Temp	14.05	degC	Excellent	41.24803	-74.8769	
32	ERMN_DEI DEWA	AdamsCre	2022-04-14T01:00:00Z	Water Temp	13.882	degC	Excellent	41.24803	-74.8769	
33	ERMN_DEI DEWA	AdamsCre	2022-04-14T02:00:00Z	Water Temp	13.69	degC	Excellent	41.24803	-74.8769	
34	ERMN_DEI DEWA	AdamsCre	2022-04-14T03:00:00Z	Water Temp	13.497	degC	Excellent	41.24803	-74.8769	
35	ERMN_DEI DEWA	AdamsCre	2022-04-14T04:00:00Z	Water Temp	13.281	degC	Excellent	41.24803	-74.8769	
36	ERMN_DEI DEWA	AdamsCre	2022-04-14T05:00:00Z	Water Temp	13.064	degC	Excellent	41.24803	-74.8769	

ERMN_DataLogger_2302125_StreamT

Ready Accessibility: Unavailable

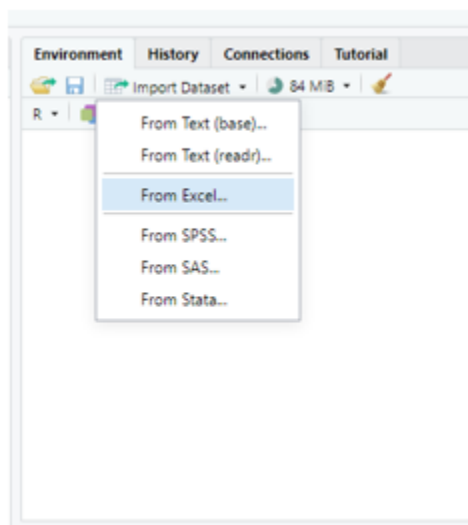
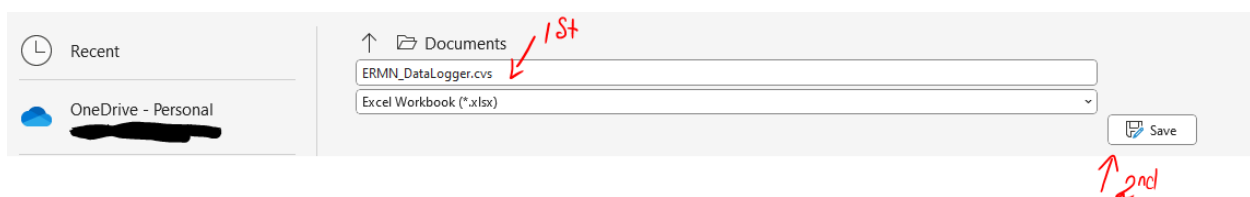
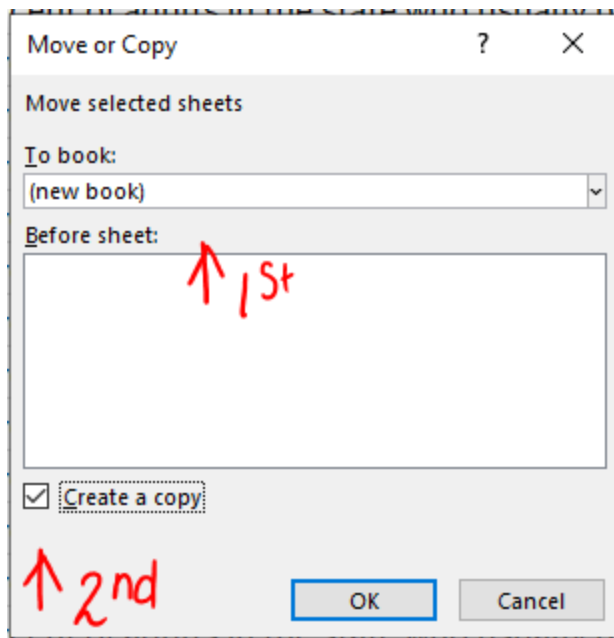
Before writing the codes in R/RStudio, let's upload the dataset into this program.

The screenshot shows the Microsoft Excel interface with the following data:

SiteID	Park	SiteName	DateTime	Variable	Value	Units	Grade	Latitude	Longitude
ERMN_DEI DEWA	AdamsCre	2022-04-12T19:00:00Z	Water Temp	11.516	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-12T20:00:00Z	Water Temp	11.88	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-12T21:00:00Z	Water Temp	12.05	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-12T22:00:00Z	Water Temp	12.195	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-12T23:00:00Z	Water Temp	12.219	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T00:00:00Z	Water Temp	12.098	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T01:00:00Z	Water Temp	11.832	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T02:00:00Z	Water Temp	11.467	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T03:00:00Z	Water Temp	11.102	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T04:00:00Z	Water Temp	10.761	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T05:00:00Z	Water Temp	10.418	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T06:00:00Z	Water Temp	10.1	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T07:00:00Z	Water Temp	9.829	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T08:00:00Z	Water Temp	9.583	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T09:00:00Z	Water Temp	9.435	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T10:00:00Z	Water Temp	9.336	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T11:00:00Z	Water Temp	9.262	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T12:00:00Z	Water Temp	9.312	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T13:00:00Z	Water Temp	9.386	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T14:00:00Z	Water Temp	9.682	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T15:00:00Z	Water Temp	10.124	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T16:00:00Z	Water Temp	10.98	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T17:00:00Z	Water Temp	12.001	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T18:00:00Z	Water Temp	12.727	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T19:00:00Z	Water Temp	13.161	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T20:00:00Z	Water Temp	13.618	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T21:00:00Z	Water Temp	13.93	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T22:00:00Z	Water Temp	14.122	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-13T23:00:00Z	Water Temp	14.146	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-14T00:00:00Z	Water Temp	14.05	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-14T01:00:00Z	Water Temp	13.882	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-14T02:00:00Z	Water Temp	13.69	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-14T03:00:00Z	Water Temp	13.497	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-14T04:00:00Z	Water Temp	13.281	degC	Excellent	41.24803	-74.8769	
ERMN_DEI DEWA	AdamsCre	2022-04-14T05:00:00Z	Water Temp	13.064	degC	Excellent	41.24803	-74.8769	

The right-click context menu is open over the 'SiteID' column header, showing options: Insert..., Delete, Rename, Move or Copy..., View Code, Protect Sheet..., Tab Color, Hide, Unhide..., Select All Sheets, and Link to this Sheet.

↑ Right-Click



Import Excel Data

File/URL: Browse...

Data Preview:

Import Options:

Name: Max Rows: ☒ First Row as Names

Sheet: Skip: ☒ Open Data Viewer

Range: NA:

Code Preview:

```
library(readxl)
dataset <- read_excel(NULL)
View(dataset)
```

Reading Excel files using readxl Import Cancel

Choose File

This PC > Documents

Organize New folder

Name	Date modified	Type	Size
ERMN_DataLogger.csv	6/18/2025 3:53 PM	Microsoft Excel W...	35,097 KB
.Rhistory	6/18/2025 11:51 AM	RHISTORY File	4 KB
ACS_physical_activity.csv	6/18/2025 9:19 AM	Microsoft Excel W...	72 KB
Workbook1	6/16/2025 8:02 PM	Microsoft Excel W...	47 KB
...	...	...	...

File name: All Files (*.*) Open Cancel

Import Excel Data

File/URL:
~/ERMN_DataLogger.cvs.xlsx Browse...

Data Preview:

SiteID (character)	Park (character)	SiteName (character)	DateTime (character)	Variable (character)	Value (double)	Units (character)	Grade (character)	Latitude (double)	Longitude (double)
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T19:00:00Z	Water Temp	11.516	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T20:00:00Z	Water Temp	11.880	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T21:00:00Z	Water Temp	12.050	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T22:00:00Z	Water Temp	12.195	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T23:00:00Z	Water Temp	12.219	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-13T00:00:00Z	Water Temp	12.098	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-13T01:00:00Z	Water Temp	11.832	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-13T02:00:00Z	Water Temp	11.467	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-13T03:00:00Z	Water Temp	11.102	degC	Excellent	41.24803	
ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-13T04:00:00Z	Water Temp	10.761	degC	Excellent	41.24803	

Previewing first 50 entries.

Import Options:

Name: Max Rows: ☒ First Row as Names

Sheet: Skip: ☒ Open Data Viewer

Range: NA:

Code Preview:

```
library(readxl)
ERMN_DataLogger_cvs_ <-
read_excel("ERMN_DataLogger
.cvs.xlsx")
View(ERMN_DataLogger_cvs_)
```

Import Cancel

[? Reading Excel files using readxl](#)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

ERMN_DataLogger_cvs_

SiteID	Park	SiteName	DateTime	Variable	Value	Units	Grade
1	ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T19:00:00Z	Water Temp	11.516	degC
2	ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T20:00:00Z	Water Temp	11.880	degC
3	ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T21:00:00Z	Water Temp	12.050	degC
4	ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T22:00:00Z	Water Temp	12.195	degC
5	ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-12T23:00:00Z	Water Temp	12.219	degC
6	ERMN_DEWA_3011	DEWA	AdamsCreek14	2022-04-13T00:00:00Z	Water Temp	12.098	degC

Showing 1 to 6 of 816,752 entries, 10 total columns

Console

```
R • R 4.5.1 • ~/
> library(readxl)
> ERMN_DataLogger_cvs_ <- read_excel("ERMN_DataLogger.cvs.xlsx")
> View(ERMN_DataLogger_cvs_)
>
```

Environment History Connections Tutorial

R • Global Environment

Data

ERMN_DataL... 816752 obs. of 10 variables

Files Plots Packages Help Viewer Presentation

Zoom Export

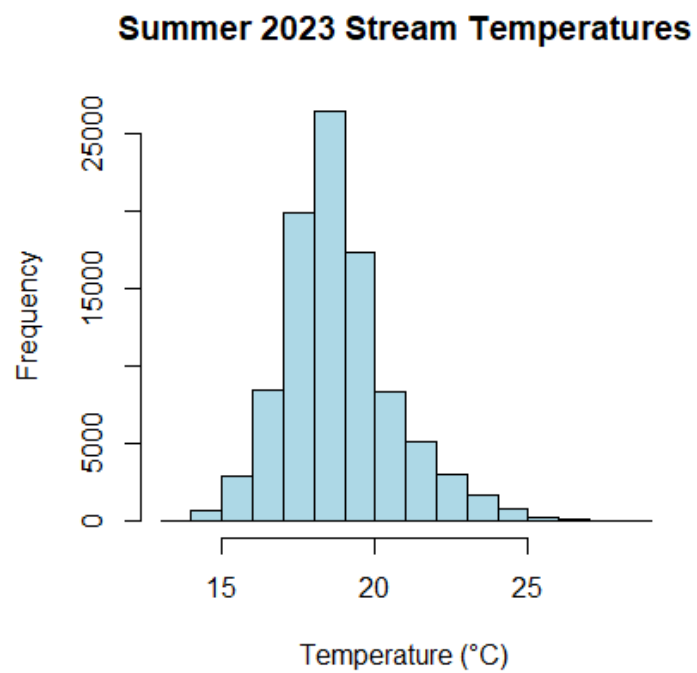
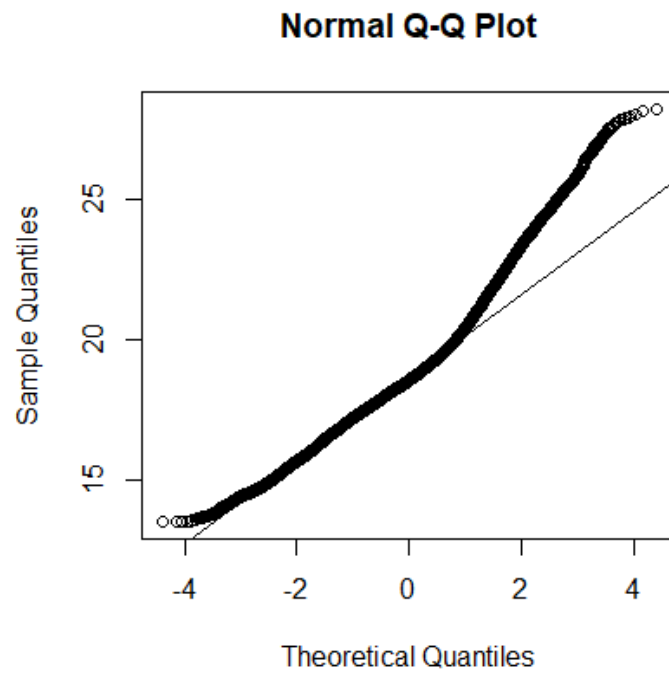
Now we will use R/RStudio to calculate the sample size, mean, and standard deviation. As well as, the t-test output, QQ plot + histogram.

```
Console Terminal Background Jobs
R 4.5.1 ~ /
> colnames(ERMN_DataLogger_cvs)
[1] "SiteID" "Park" "SiteName"
[4] "DateTime" "Variable" "Value"
[7] "Units" "Grade" "Latitude"
[10] "Longitude"
> str(ERMN_DataLogger_cvs)
tibble [816,752 × 10] (S3: tbl_df/tbl/data.frame)
 $ SiteID : chr [1:816752] "ERMN_DEWA_3011" "ERMN_DEWA_3011" "ERMN_DEWA_3011" "ERMN_DEWA_3011" ...
 $ Park : chr [1:816752] "DEWA" "DEWA" "DEWA" "DEWA" ...
 $ SiteName : chr [1:816752] "AdamsCreek14" "AdamsCreek14" "AdamsCreek14" "AdamsCreek14" ...
 $ DateTime : chr [1:816752] "2022-04-12T19:00:00Z" "2022-04-12T20:00:00Z" "2022-04-12T21:00:00Z" "2022-04-12T22:00:00Z" ...
 $ Variable : chr [1:816752] "Water Temp" "Water Temp" "Water Temp" "Water Temp" ...
 $ Value : num [1:816752] 11.5 11.9 12.1 12.2 12.2 ...
 $ Units : chr [1:816752] "degC" "degC" "degC" "degC" ...
 $ Grade : chr [1:816752] "Excellent" "Excellent" "Excellent" "Excellent" ...
 $ Latitude : num [1:816752] 41.2 41.2 41.2 41.2 41.2 ...
 $ Longitude: num [1:816752] -74.9 -74.9 -74.9 -74.9 -74.9 ...
>
> ERMN_DataLogger_cvs$DateTime <- as.POSIXct(ERMN_DataLogger_cvs$DateTime, format="%Y-%m-%dT%H:%M:%OSZ", tz="UTC")
>
> df_temp <- subset(ERMN_DataLogger_cvs, Variable == "Water Temp")
>
```

```
Console Terminal Background Jobs
R 4.5.1 ~ /
> df_summer <- subset(df_temp, DateTime >= as.POSIXct("2023-07-01", tz="UTC")
&
+           DateTime < as.POSIXct("2023-09-01", tz="UTC"))
>
> temps <- df_summer$Value
>
> length(temps)      # sample size
[1] 94827
> summary(temps)     # summary stats
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  13.52  17.65   18.56   18.80   19.65   28.22
> sd(temps)          # sample sd
[1] 1.803826
>
> hist(temps, main="Summer 2023 Stream Temperatures", xlab="Temperature (°C)",
col="lightblue")
> qqnorm(temps); qqline(temps)
>
> t.test(temps, mu = 18, conf.level = 0.95)

      One Sample t-test

data:  temps
t = 136.56, df = 94826, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 18
95 percent confidence interval:
 18.78846 18.81142
sample estimates:
mean of x
 18.79994
```



Based on the results:

The test statistic is 136.56

The degrees of freedom is 94,826

The sample mean is 18.80°C

The 95% confidence interval is $(18.78846^{\circ}\text{C}, 18.81142)$ degrees of Celsius

We are 95% confident the population mean of summer 2023 stream temperatures is between 18.78846°C and 18.81142°C .

This implies that in about 95% of all samples of summer 2023 stream temperatures, the confidence interval will contain the true population mean within the interval $(18.79^{\circ}\text{C}, 18.81^{\circ}\text{C})$.

References (APA 7 Format)

- Centers for Disease Control and Prevention. (n.d.). *Nutrition, physical activity, and obesity, American Community Survey* [Data set]. Data.gov. <https://catalog.data.gov/dataset/nutrition-physical-activity-and-obesity-american-community-survey>
- Chukwuemeka, S. D. (2023). *Inferential statistics: Projects*. Statistical Science. <https://statistical-science.appspot.com/InferentialStatistics/Projects.html#projectRequirements>
- RStudio Team. (2023). *RStudio: Integrated development environment for R* (Version 2023.06.1) [Computer software]. RStudio, PBC. <https://www.rstudio.com/>
- U.S. Department of the Interior, National Park Service. (n.d.). *Sub-hourly stream temperature measurements throughout Delaware Water Gap National Recreation Area* [Data set]. Data.gov. <https://catalog.data.gov/dataset/sub-hourly-stream-temperature-measurements-throughout-delaware-water-gap-national-recreati>